

Mihir A. Kulkarni

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EDUCATION

Rutgers University, New Brunswick

Master of Science in Computer Science

Aug 2024 – May 2026

New Brunswick, NJ

Savitribai Phule Pune University

Bachelor of Technology in Computer Engineering

Aug 2019 – May 2023

Pune, India

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS

Technologies: OpenGL, HuggingFace, TensorFlow, Scikit-Learn, AWS, MySQL, PostgreSQL, ReactJS, React Native, Node.js, Express.js, MongoDB, Linux, Unix, Neovim, Vim, Bash, Object-Oriented Programming

Conceptual Areas: Algorithms, DBMS, Artificial Intelligence, Deep Neural Networks, Operating Systems, Computer Networks, REST, Git, Agile, TCP/IP, Backend Systems, Data Structures, Web Development, Object-Oriented Design, IP Networking, Databases

PROFESSIONAL EXPERIENCE

Software Engineering Intern

May 2025 – Aug 2025

UniteGPS (Transportation & Routing Solutions)

Portland, ME

- Led the development of a SaaS alarm response system (MERN full stack) for real-time contract bidding and acceptance for 50+ concurrent ADT alarm response contractors, with potential to secure \$250K investment.
- Implemented location-based validation (Expo Location library) and in-app reporting, projected to reduce contractor reporting discrepancies by 40%.
- Deployed a Node.js microservice automating contract assignment using a round-robin algorithm, eliminating manual assignment and reducing allocation time by 100%.
- Reduced crash reports on low-memory Android devices by 70% by embedding an in-app camera using the react-native-camera library, replacing unstable external camera workflows.

Software Engineer

Jun 2023 – May 2024

Syngenta Pvt. Ltd (Agri Tech MNC using AI for Digital Farming Solutions)

Pune, India

- Led a serverless data ingestion pipeline (Python, AWS Glue, Redshift) for Price Capture Dashboard, automating ETL for pricing data from 50+ global retailers via Octoparse API, reducing manual effort by 80% and enabling analytics across 20+ EAME markets.
- Improved data reliability by eliminating ~100% ingestion errors through a custom JavaScript data validation tool in Qlik Mashups, ensuring accurate ingestion into AWS Redshift.
- Accelerated dashboard load time by 4x through SQL optimization and caching lookup tables in Qlik Sense, improving stakeholder experience.
- Enabled 20+ commercial and finance stakeholders across EAME to self-serve pricing analytics by leading QlikSense dashboard onboarding and feedback sessions, reducing support tickets by ~60%

Associate Software Engineer

Aug 2022 – Jun 2023

Syngenta Pvt. Ltd

Pune, India

- Reduced pricing-approval SLA from 48 to 24 hours (60% faster) by engineering an event-driven pipeline using Qlik Sense webhooks, Power Automate, and SharePoint Lists.
- Engineered writeback functionality in Qlik Sense via Qlik Mashups, replacing costly third-party integrations and cutting licensing costs by 30%.
- Built and deployed a Python time-series forecasting module (Pandas, Prophet) adopted by 10+ finance teams, improving forecasting accuracy (MAPE) by ~22% and enabling better pricing decisions.

SELECTED PROJECTS

Isometric 2D Strategy Game Engine | C++, OpenGL 4.8, GLFW, GLM

Spring 2025

- Built an Age of Empires-style isometric renderer using instanced VBOs and a texture atlas shader, sustaining 120 FPS while rendering 10K+ tiles per frame.
- Implemented ambient and directional Phong lighting with shadow mapping, maintaining <6ms render time per frame, and modularized the codebase into ECS components for future AI and pathfinding extensions.

High-Performance Concurrent TCP Chat Server | C, POSIX Sockets, Linux epoll, Non-Blocking I/O, Docker

Summer 2025

- Developed a concurrent TCP chat server in C using epoll, supporting thousands of simultaneous client connections with non-blocking I/O and structured state management.
- Implemented backpressure handling to safely drop slow or oversized consumers, ensuring reliability under reconnect storms and heavy load.
- Integrated structured logging (log_kv) and exposed Prometheus-compatible metrics (connections_current, bytes_in/out, messages_total, dropped_clients_total) for real-time observability.
- Dockerized the server with multi-stage builds for portable deployment and designed benchmarking harnesses to measure throughput and latency, achieving sub-3 ms p95 latency under moderate load.

Grid-World AI Agent | Python, PyTorch, NumPy

Fall 2024

- Built two AI agents for fire extinguishing and rat tracking using A* pathfinding, Bayesian inference, and probability-based heuristics, cutting target localization time by 50%.
- Engineered a hybrid CNN-FCN model for time-step forecasting in sequential decision tasks, achieving 88% prediction accuracy.